Plasma total homocysteine status of vegetarians compared with omnivores: a systematic review and meta-analysis.

There is strong evidence indicating that elevated plasma total homocysteine (tHcy) levels are a major independent biomarker and/or a contributor to chronic conditions, such as CVD. A deficiency of vitamin B■■ can elevate homocysteine. Vegetarians are a group of the population who are potentially at greater risk of vitamin B■■ deficiency than omnivores. This is the first systematic review and meta-analysis to appraise a range of studies that compared the homocysteine and vitamin B■■ levels of vegetarians and omnivores. The search methods employed identified 443 entries, from which, by screening using set inclusion and exclusion criteria, six eligible cohort case studies and eleven cross-sectional studies from 1999 to 2010 were revealed, which compared concentrations of plasma tHcy and serum vitamin B■■ of omnivores, lactovegetarians or lacto-ovovegetarians and vegans. Of the identified seventeen studies (3230 participants), only two studies reported that vegan concentrations of plasma tHcy and serum vitamin B■■ did not differ from omnivores. The present study confirmed that an inverse relationship exists between plasma tHcy and serum vitamin B■■, from which it can be concluded that the usual dietary source of vitamin B■■ is animal products and those who choose to omit or restrict these products are destined to become vitamin B■■ deficient. At present, the available supplement, which is usually used for fortification of food, is the unreliable cyanocobalamin. A well-designed study is needed to investigate a reliable and suitable supplement to normalise the elevated plasma tHcy of a high majority of vegetarians. This would fill the gaps in the present nutritional scientific knowledge.